Pothuri Rahul

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.02
Alphores Junior College	Class XII (Telangana Board of Secondary Education)	2024	97.10%
Pragnya Global School	Class X (CBSE)	2022	85.80%

RELEVANT COURSES

Core Courses : Signal Processing | Digital Systems | Semiconductor Devices | Analog Electronics | Circuit Analysis | Control Systems | Engineering Electromagnetics | Electrical Machines | Power System Analysis | Machine Learning

Lab Courses : Digital Systems Lab | Electronic Devices Lab | DSP Lab | PCB Design & Layout | Electrical Circuits Lab

Programming : Scientific Programming for Electrical Engineers | Data Structures & Applications | Introduction to Programming

Math : Probability | Matrix Theory | Differential Equations | Maths for EE | Vector Calculus | Random Processes

Online Certified Courses : Signal Processing On-ramp by MATLAB , Core Signal Processing Techniques in MATLAB

SKILLS

Technical : C , Python , SQL , Matlab

Audio & DSP : FFT Analysis , MIR , Audio Signal Processing

Collaboration & Communication : Pitching, Negotiations, Collaborating

PROJECTS

Microphone Amplifier with class AB Driver :

- Designed a Microphone Audio Amplifier chain (Microphone -> Pre Amp -> Class AB Driver -> Speakers) after researching on how the speaker get drived by the amplifier.
- Performed **analog circuit design, LTspice simulation, and hardware validation** of the amplifier.
- The outcome of this amplifier is Scaled Audio signal, with least noise and distortion.
- Experimentally characterized amplifier performance through **THD (FFT analysis), PSRR, noise measurements, bandwidth testing, and output power validation** using laboratory instruments including oscilloscopes and function generators.

5 bit Analog to Digital converter :

- Built an Analog - Digital Converter using Successive Approximation Register (SAR) algorithm.
- Constructed an R-2R ladder Digital-to-Analog Converter for reference voltage generation. Implemented SAR control logic using Arduino. Used OpAmp for the R-2R Summing amplifier.
- Developed the **SAR control logic** for successive approximation and digital code generation.

FFT-Based Musical Note and Chord Detection :

- Developed an **FFT-based musical note and chord detection system** for monophonic and polyphonic audio signals.
- Implemented **spectral peak detection, MIDI note mapping, and beat-wise chord analysis** from scratch.
- Improved chord estimation by **filtering harmonic frequencies** and refining spectral peak selection.
- **Implemented entirely using classical Digital Signal Processing techniques (no Machine Learning/Deep**

Learning) to understand the complete signal processing pipeline.
- Evaluated the algorithm on recorded audio signals using **Python, NumPy, SciPy, and Librosa**.

Audio Fingerprinting and Music Identification System :

- Implemented an **audio fingerprinting system** (Shazam type Prototype)using **FFT, spectral peak detection, and constellation map generation**.
- Generated compact **hash fingerprints** from spectral features for audio identification.
- Built an optimized **SQLite-based fingerprint database** for scalable music indexing and retrieval. This system is working for Small pieces for original Songs and for also Cover songs sang on karaoke tracks of original one
- Developed a **time-offset voting algorithm** for accurate song detection from short audio clips.

Digital Signal Processing Experiments

Fixed-Point Division using Newton-Raphson

- Implemented a **Q16 fixed-point divider** using the Newton-Raphson reciprocal algorithm.
- Designed **normalization**, **LUT-based initial approximation**, and verified the implementation in **MATLAB** and **Verilog HDL**.

FIR Filter Design & Hardware Implementation

- Designed a **100-tap low-pass FIR filter** in MATLAB and implemented it in **Verilog HDL**.
- Compared **Direct Form, Generate-based, Symmetric, and Pipelined FIR architectures** for functionality and hardware implementation.

EXTRACURRICULAR

- NSS Volunteer
- Actively Participating in Student Activities
- Listening to Music